Eva Abou Harb

Professor Esmali

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Lab Reflection: GitHub and Jupyter Notebook Exploration

During this lab, I learned the basics of GitHub and Jupyter Notebooks, which are two important tools used in computer science and artificial intelligence projects. I started by logging into my GitHub account and exploring the interface to become familiar with repositories, commits, and the overall layout of the platform.

Next, I created a new public repository called jupyter-exploration and initialized it with a README file. I edited the README by adding a brief description of the purpose of the repository and then committed the changes directly to the main branch.

For the Jupyter Notebook portion of the lab, I used Google Colab because it already includes a Jupyter environment and does not require any software installation. I created a new notebook named **My_First_Notebook.ipynb**. Inside the notebook, I added a Markdown cell with a short message and a Python code cell containing the command print("Hello, World!"). After creating the cells, I executed them using the play button on the left side of each cell and verified that the output appeared correctly.

One part of this lab that was new for me was using GitHub Desktop. In previous classes, I usually created the repository online and uploaded files directly through the GitHub website. This time, I followed the instructions to clone my repository using GitHub Desktop, copied my notebook into the local repository folder, committed the changes, and pushed them back to

GitHub. After the upload was complete, I verified that my notebook was successfully stored in my online repository.

This lab helped me better understand the purpose of version control and why GitHub is such an important platform for developers and AI professionals. Before this assignment, I mainly thought of GitHub as a place to store files, but I learned that it also tracks changes over time through commits, making it easier to organize projects and collaborate with others.

I also learned more about Jupyter Notebooks and why they are widely used in data science, machine learning, and computer vision. I like that they allow code, explanations, and outputs to exist in the same document, making experiments easier to organize and understand.

The biggest new concept for me was working with GitHub Desktop. At first, I was more comfortable uploading files directly through the GitHub website because that was the method I had always used. However, after completing the cloning, committing, and pushing process, I realized that GitHub Desktop is actually very convenient and provides a better workflow for managing projects. I now understand the relationship between a local repository and the online repository stored on GitHub.

I enjoyed this lab because it introduced me to tools that are commonly used in real-world software development and AI projects. Although I had previous experience with GitHub, learning how to use GitHub Desktop was a valuable addition to my skill set. I also appreciated using Google Colab because it made working with Jupyter Notebooks more convenient and accessible. Overall, I think this was a very useful introduction to the technologies that will be used throughout the course.